KOLHAPUR INSTITUTE OF TECHNOLOGY'S

COLLEGE OF ENGINEERING, KOLHAPUR

(EMPOWERED AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

Mini Project-I | Course Code: UCBIL0371

TRUSTFIX – TRUSTED HOME SERVICE PLATFORM

Synopsis Report

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Academic Year: 2026–27 | Semester: III

Submission Date: _____

HOD: _____

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Abstract

TrustFix – Trusted Home Service Platform is a web-based mini-project intended to make it easier for residents of Kolhapur to discover and book local home-service professionals. The initial service scope focuses on three common categories: Electrician, Plumber and AC Repairer. In the conventional approach, customers often depend on personal references, local contacts or repeated searches to find a suitable professional. Information about availability, service experience, pricing and previous customer feedback may be scattered or difficult to compare. Booking details may also be exchanged informally, making it harder for customers and providers to maintain a clear record of requests.

TrustFix proposes a common digital platform connecting three user roles: Customer, Service Provider and Administrator. Customers can access the available service categories, view provider information and submit service-booking requests. Service providers can view and manage incoming requests through a provider-facing interface. Administrators have a separate dashboard intended to support the management of users, providers, service categories, bookings and reviews. The intended workflow provides a

structured way to move a booking from request to completion and to collect customer feedback after service delivery.

The project is partly completed. The team has identified working modules including login, service browsing, booking, an administrator dashboard, a provider dashboard and reviews. A payment interface or flow is available in demo form; it must not be represented as a live payment-gateway integration or as real-money transaction processing. Screenshots and source-code evidence are to be added after they are supplied and checked.

The earlier TrustFix project discussions identified React.js for the frontend, Java Spring Boot for backend services and MySQL for data storage. this report treats React.js, Spring Boot and MySQL as the previously identified implementation direction, subject to verification against the team's current source code and dependency files. The project is an academic mini-project and is not claimed to be a production-ready service. Its purpose is to demonstrate the design and partial implementation of a role-based local-service booking application.

1. Introduction

Home maintenance and repair services are recurring requirements in residential areas. Electrical faults, plumbing problems and air-conditioning maintenance often require customers to locate a professional who is suitable for the particular task. In many local settings, customers identify service providers through friends, neighbours, shop contacts or informal online searches. Although these methods can work, the information available to the customer is not always organized in one place.

A customer may need to contact several professionals to ask about the type of service offered, approximate charges, location and availability. Details can be repeated across phone calls or messaging conversations. After a request is made, it may be difficult to track whether the provider has accepted it, when the work is expected to begin or whether the service has been completed. Providers may also find it difficult to manage several customer enquiries without a central record.

TrustFix is proposed as a web-based platform to organize this interaction. It focuses on Kolhapur and initially includes Electrician, Plumber and AC Repairer categories. The application separates access according to user role so that customers, service providers and administrators are presented with functions relevant to their responsibilities.

The customer side is intended to support account access, service discovery, provider selection, booking and feedback. The service-provider side is intended to support provider information and booking-request handling. The administrator side is intended to provide oversight of the platform's main records. The exact availability of each operation depends on the current implementation; this report records the modules identified by the team as implemented while avoiding claims about unverified sub-features.

The project is being developed incrementally as a Mini Project-I. The current status is partly completed. The report therefore describes the problem, motivation, objectives and planned system behaviour while clearly identifying the limits of the current build. Detailed implementation evidence, screenshots, test cases and final results can be added in a later revision after the corresponding application screens and code are reviewed.

2. Motivation

The main motivation for TrustFix is the everyday difficulty of finding and coordinating a suitable homeservice professional. Customers may have to rely on word-of-mouth recommendations or contact multiple providers individually. This can consume time, especially when the customer needs a specific service such as electrical repair, plumbing work or AC servicing.

A second motivation is the lack of a consistent booking record in informal service arrangements. When requests are handled through separate calls and messages, customers and providers may not have a shared view of the request details or its current status. A structured booking record can make the interaction easier to follow and can reduce confusion about whether a request is pending, accepted, in progress or completed.

Trust and service quality are also important considerations. Customers often want some indication of a provider's experience or previous customer feedback before making a choice. A platform that supports provider profiles and ratings or reviews can make relevant information easier to find. Reviews do not guarantee service quality, but they can provide additional context for a customer's decision.

The project also provides the student team with practical experience in building a multi-role software application. It brings together frontend development, backend application logic, database operations, authentication, role-based access, form handling and the management of related records. These are useful concepts for understanding how a real software application is divided into components and how those components communicate.

The scope has deliberately been kept suitable for a college mini-project. The first version concentrates on a limited geographic area and three service categories instead of attempting to cover every type of home service. Payment is currently treated as a demonstration feature, not a confirmed live financial integration. Advanced production concerns such as large-scale provider verification, continuous location tracking, complex payment settlement and high-availability infrastructure are outside the current academic scope.

3. Objectives

The primary objective of TrustFix is to design and partially implement a web-based platform that organizes the discovery and booking of selected home-service professionals in Kolhapur.

The specific objectives are:

1. To provide separate access and interfaces for Customers, Service Providers and Administrators, according to their roles.
2. To organize the initial service categories—Electrician, Plumber and AC Repairer—so that customers can find services through a common platform.
3. To provide a customer-facing workflow for selecting a service and submitting a booking request to a service provider.
4. To provide a provider-facing dashboard for viewing and handling service requests, subject to the functions supported by the current implementation.
5. To provide an administrator dashboard for overseeing platform information and managing relevant records.
6. To support ratings and reviews so that customers can share feedback about service experiences where the review workflow is available.
7. To demonstrate a payment-related user flow for academic purposes while clearly distinguishing a simulated/demo payment from a live payment gateway.
8. To organize application data using a database and maintain appropriate relationships between users, providers, services, bookings and reviews.
9. To apply software engineering practices such as modular development, input validation, role separation, debugging and iterative testing.
10. To identify limitations in the current partly completed build and define realistic future improvements without claiming that unimplemented features are already available.

4. Problem Statement

In Kolhapur, customers who require household repair or maintenance services may find professionals through personal contacts, recommendations or scattered online information. This process can require repeated enquiries and does not always provide a consistent way to compare providers or maintain service request details. Customers may not have a convenient shared record of the provider selected, the service requested or the progress of the booking.

Service providers may receive requests through different communication channels and may need to keep track of customer details and pending work manually. Without a structured request-management interface, it can become difficult to review incoming requests and update customers about their status. Platform administrators, where applicable, also need a central way to manage users, providers, service categories, bookings and customer feedback.

Therefore, there is a need for a simple digital platform that brings customers and local service providers together, organizes the selected service categories, supports booking requests and provides role-specific interfaces for customers, providers and administrators. TrustFix is proposed to address this need for a limited set of home services in Kolhapur.

The problem addressed by this mini-project can be stated as follows: to design and partly implement a

role-based web application that simplifies the discovery and booking of local electricians, plumbers and AC repairers, provides a structured interface for service providers to handle requests, and enables administrators to manage key platform information. The current build is partly complete, and its payment functionality is demonstrative rather than a verified live payment service.

5. Proposed System and Scope

TrustFix follows a three-role concept. A Customer uses the platform to access service categories, review available provider information and create a booking. A Service Provider uses a separate provider-facing interface to view or manage requests. An Administrator uses an administrative interface to manage platform records. The purpose of separating these roles is to keep each user's available actions relevant to their responsibilities.

The initial service categories are Electrician, Plumber and AC Repairer. The project is limited to the Kolhapur service area for the Mini Project-I scope. The application is intended to be accessed through a web browser. The precise deployment environment and supported browser matrix have not yet been documented and should be recorded after testing.

The team has identified the following modules as implemented or available in the current partly completed build: login, service browsing, booking, administrator dashboard, provider dashboard, reviews and a demo payment flow. This status statement records the team's report; it is not an independent code audit. The precise sub-functions, validation rules, persistence behaviour and role restrictions should be confirmed through screenshots, source-code inspection and test execution before final submission.

The earlier project discussions identified React.js, Java Spring Boot and MySQL as the working technology direction. The original SRS proposed HTML, CSS, JavaScript and Node.js. Since the user has indicated that the actual technology stack differs from the original plan, the team must verify the exact stack against the current project files. No unconfirmed libraries, hosting services, payment providers or deployment tools are named in this report.

The current scope excludes claims of real-time GPS tracking, automated provider background verification, AI recommendations, dedicated Android/iOS applications, multilingual support and production-grade payment processing. These may be considered only as future enhancements if time, requirements and implementation resources permit.

6. Current Implementation Status

The project is partly completed. The following status is based on the team's supplied update and should be revised if subsequent code review shows that a listed feature is only a static screen or prototype.

- Login: identified by the team as implemented. The exact authentication mechanism, passwordstorageapproach, session/token handling and validation behaviour must be confirmed from the source code.

- Service browsing: identified as implemented for the service-discovery experience. The final report should confirm whether the category and provider information is loaded dynamically from the backend or displayed as fixed/demo data.

- Booking: identified as implemented. The team should verify whether booking records are persisted, whether providers can update request status and which booking states are supported.

- Provider dashboard: identified as implemented. The exact provider actions and their effect on stored booking data should be demonstrated.

- Administrator dashboard: identified as implemented. The exact management operations available to the administrator should be listed after checking the current screens and backend endpoints.

- Reviews: identified as implemented. The team should confirm whether reviews are linked to a particular booking and whether submission is restricted to customers who completed a service.

- Payment: available as a demo. It must not be described as a real payment-gateway integration, successful financial settlement or production payment facility unless a real provider integration and transaction verification are subsequently demonstrated.

Screenshots and source-code extracts have not yet been supplied for this report. Consequently, no screenshot numbers, code listings, measured performance values, test pass rates or user-evaluation results are fabricated here. Those items should be added after the team provides the evidence and the relevant tests have actually been run.

7. References

[1] TrustFix Team, “TrustFix – Trusted Home Service Platform: Software Requirements Specification (SRS),” Mini Project-I, Department of Computer Science and Business Systems, KIT’s College of Engineering, Kolhapur, 2026–27.

[2] React, “React Documentation.” Available: https://react.dev/

[3] VMware Tanzu / Spring, “Spring Boot Reference Documentation.” Available: https://docs.spring.io/spring-boot/index.html

[4] Oracle Corporation, “MySQL Reference Manual.” Available: https://dev.mysql.com/doc/

[5] Mozilla, “MDN Web Docs: HTML, CSS and JavaScript.” Available: https://developer.mozilla.org/

[6] OWASP Foundation, “OWASP Top Ten Web Application Security Risks.” Available: https://owasp.org/www-project-top-ten/

[7] World Wide Web Consortium (W3C), “Web Standards.” Available: https://www.w3.org/

Note: References [2]–[4] describe the technology direction previously identified for the working project and must be retained only if those technologies are confirmed in the submitted source code. If the actual stack differs, replace these entries with the official documentation for the technologies actually used.